A3-PS003-TC188

Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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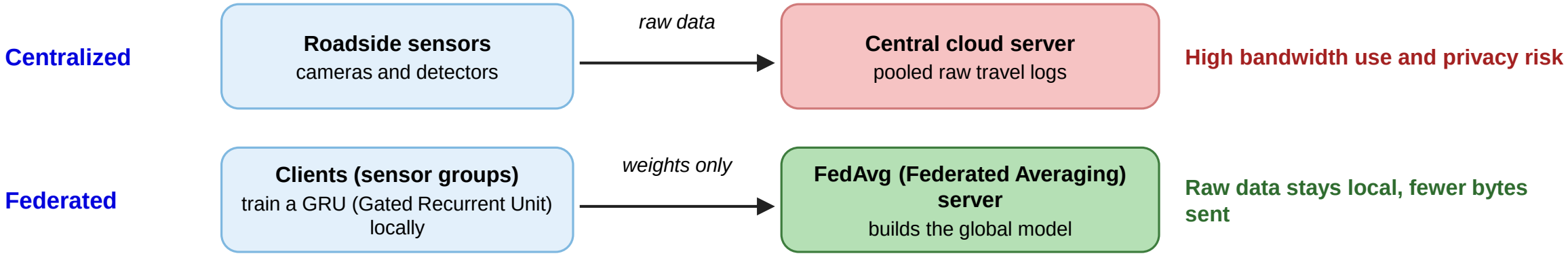
AIORI Hackathon 3

October 11, 2026

Problem Description

- Urban traffic congestion causes delays, higher emissions and increased operational costs.
- Traffic forecasting enables better traffic management and planning.
- Centralized training requires sharing data from multiple locations, which raises privacy and data ownership concerns.
- Traffic data across regions follows different patterns and is often non IID (not Independent and Identically Distributed).
- The problem is to build an accurate and privacy preserving traffic forecasting model using federated learning on a real world dataset.

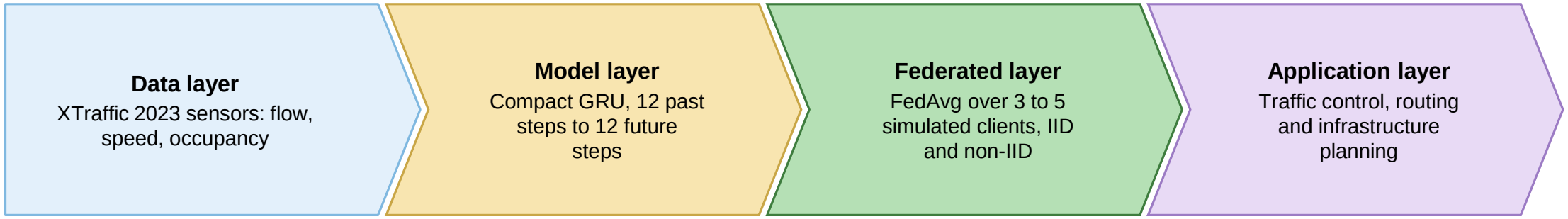
Centralized vs federated data flow



Background

- Intelligent transportation systems rely on short term traffic forecasting for traffic control, routing and infrastructure planning.
- Deep learning models such as GRU (Gated Recurrent Unit) and LSTM (Long Short-Term Memory) have shown strong performance in time series forecasting.
- Federated learning enables collaboration across multiple data owners without sharing raw data.
- Real world traffic data is naturally heterogeneous due to geographic, road type and local events, making it suitable for evaluating IID (Independent and Identically Distributed) and non IID scenarios.
- The XTraffic dataset provides large scale real world traffic data with geographic information, which is well suited for this project [8].

Layered view of the solution



Literature & RFC Review

Sl. no	Technique	Concept	Measurements	Numerical data	Future enhancements
1	**FedAvg** McMahan et al., 2017 [1]	Clients run local SGD (Stochastic Gradient Descent); the server averages their weights, weighted by data size, every round.	Communication rounds to reach a target accuracy	10 to 100 times fewer rounds than synchronized SGD	Client selection and proximal terms for non-IID data
2	**Centralized GRU baseline** Benchmark in [2]	Gated recurrent network on the last 12 flow readings, trained on pooled sensor data.	MAE (Mean Absolute Error), RMSE (Root Mean Square Error), MAPE (Mean Absolute Percentage Error), R^2 at 15 to 60 min	MAE 25.60, RMSE 42.14 (LargeST, District 12)	Move to federated training so raw data stays local
3	**Fed-GDAN** Li et al., 2025 [3]	Federated graph diffusion attention network with adaptive local aggregation and differential privacy.	MAE, RMSE, MAPE, R^2	MAE 30.44, RMSE 46.15 (re-run in [2])	Lighter model to cut the full-weight upload per round
4	**FedLLM** Kaur et al., 2026 [2]	Qwen2.5-1.5B tuned with LoRA (Low-Rank Adaptation); only the adapters are averaged by FedAvg across 4 non-IID clients.	MAE, RMSE, R^2, MB per round	MAE 15.07, RMSE 23.31, R^2 0.985; 70.5 MB per round vs 2.9 GB full model	Traffic-aware aggregation, more clients and rounds
5	**FedTraffic** Abboud and Khalil, 2026 [4]	Edge-Fog-Cloud federation with a TCN-CVAE (Temporal Convolutional Network - Conditional Variational Autoencoder) local model, clustering, fuzzy client selection and FedProx (Federated Proximal).	MAE, RMSE, MAPE, R^2, communication overhead	MAE 2.12, RMSE 4.28, MAPE 5.47%, R^2 0.966; 8:1 latent compression	Cluster sensors by traffic behaviour to build non-IID clients

From the literature to our approach

Centralized GRU / GNN (Graph Neural Network)
Accurate, but raw sensor data is pooled on one server [2]

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FedAvg and federated GNNs
Raw data stays local, but full weights are uploaded [1][3]

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Open gap
Non-IID drift and bytes per round need joint evaluation [5][7]

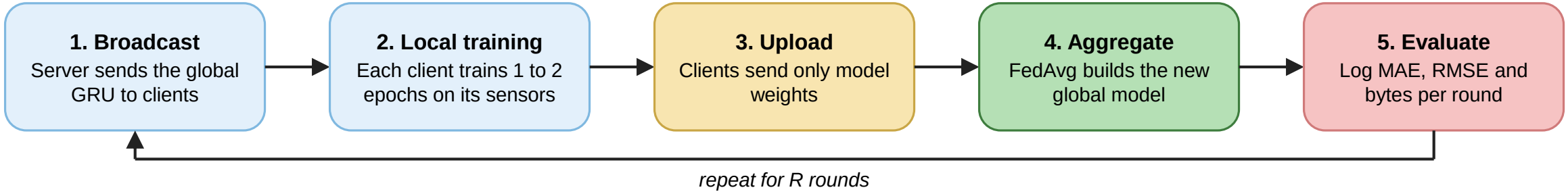
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Our approach
Compact GRU + FedAvg, IID vs non-IID, MAE/RMSE and bytes per round

Proposed Solution

- Use the XTraffic dataset containing traffic flow, speed and occupancy for the year 2023.
- Partition sensors into 3 to 5 simulated clients.
- Use a GRU based model for multi-step traffic forecasting (12 past steps to 12 future steps).
- Train a global model using Federated Averaging (FedAvg) where each client trains locally and shares model weights.
- Evaluate the model in IID and non IID settings.
- Compare with centralized training and local only training.
- Use MAE and RMSE for accuracy and measure communication bytes per round and convergence.

One federated training round



Optimization Proposed

- Use a compact GRU model to reduce communication cost while maintaining good accuracy.
- Apply data normalization and sliding window to prepare input sequences.
- Use client selection and limited local epochs to speed up training.
- Experiment with different client partitioning strategies for IID and non IID scenarios.
- Evaluate the impact of model size and number of clients on accuracy and communication cost.

Aspect	Baseline	Optimized Approach
Model	LSTM	Compact GRU
Clients	All clients every round	Client selection (fraction)
Local epochs	5	1 to 2
Input window	Fixed, no scaling	Normalized, 12 steps
Data split	Random split	Chronological split
Communication	Higher (larger model)	Lower (smaller model)
Performance	Baseline accuracy	Higher accuracy, faster convergence

Solution Architecture & Design

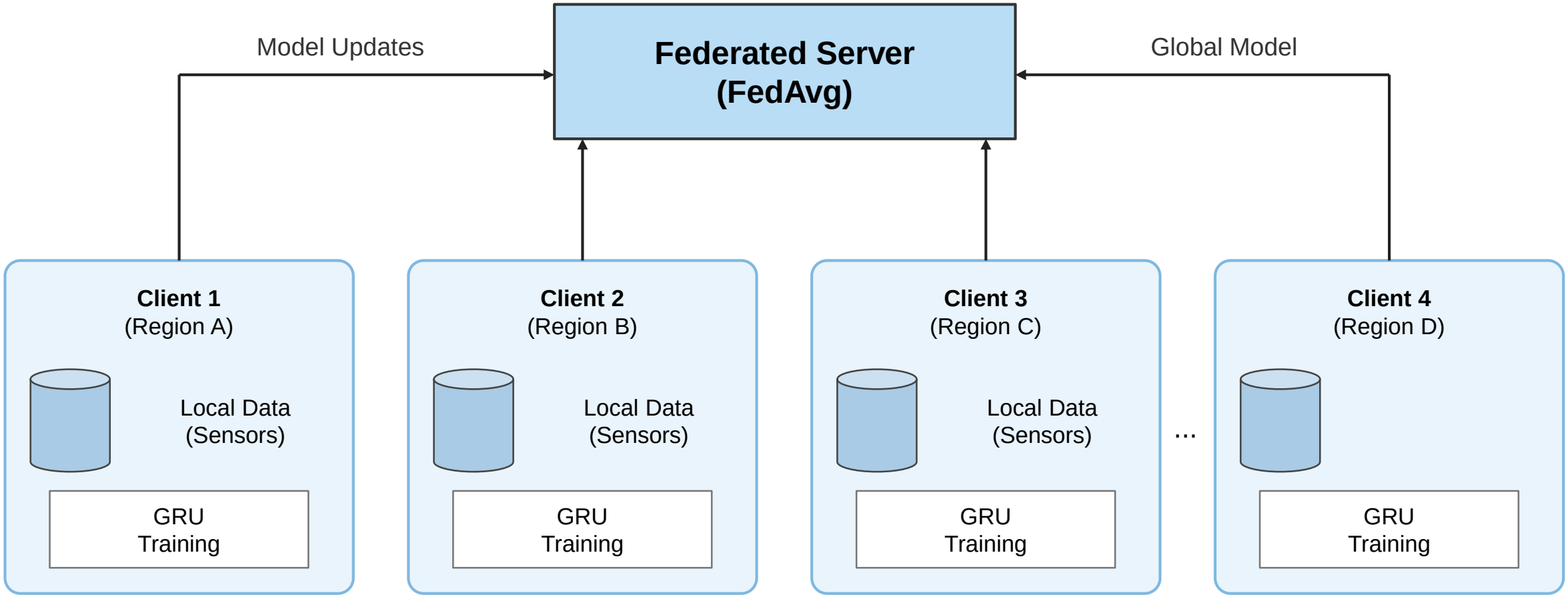


Figure: System Architecture UML Diagram

Data Flow Diagram

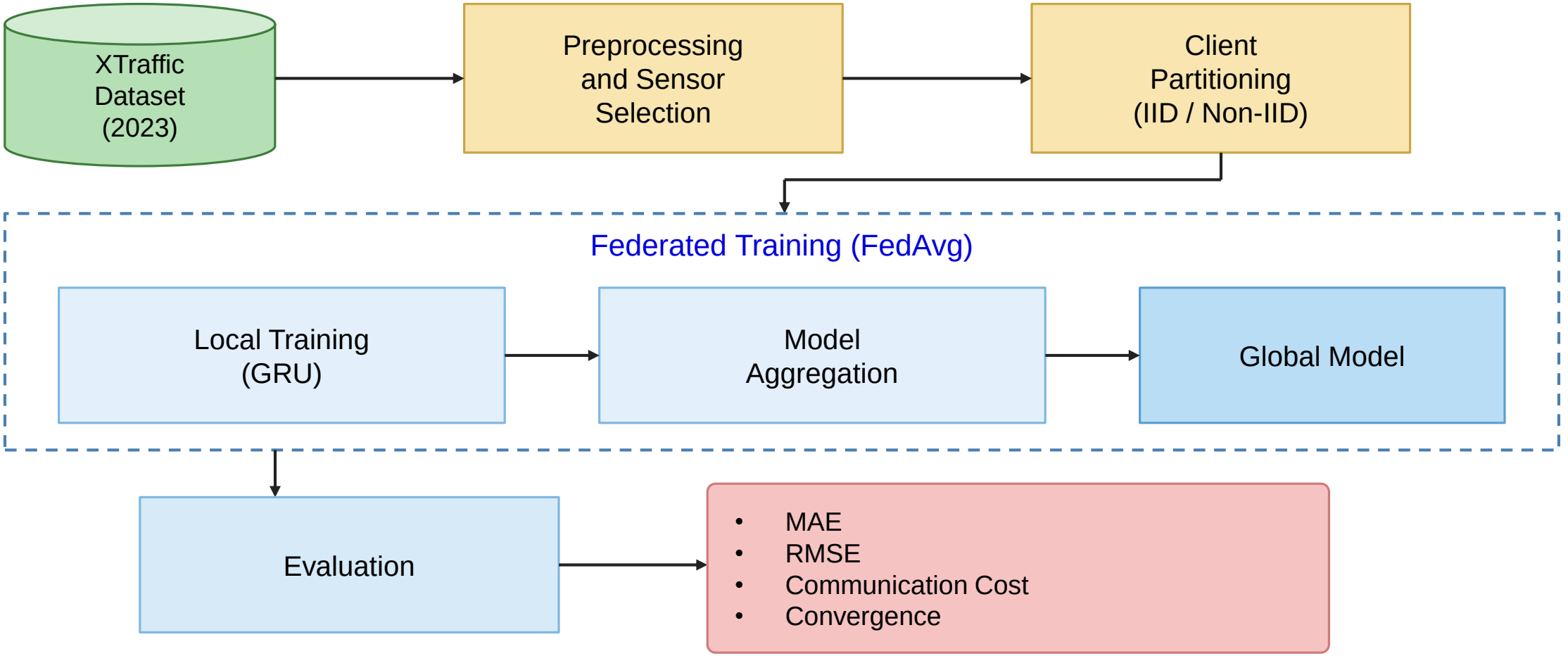


Figure: Data Flow Diagram (DFD)

Deployment Timeline

M1 Literature Review
Survey of GRU, FedAvg and non IID studies

M2 Dataset Download and Analysis
XTraffic 2023 flow, speed and occupancy explored

M3 Data Preprocessing
Normalization, 12-step windows, chronological split

M4 Model Development (GRU)
Compact GRU for 12-step forecasting

M5 Federated Learning Implementation
FedAvg server, 3 to 5 simulated clients

M6 Experiments (IID and Non-IID)
Runs with both client partitioning strategies

M7 Evaluation and Analysis
MAE, RMSE, communication cost, convergence

M8 Report and Presentation
Final report, results and slides

Methodology

- Milestones M1 to M8 run in sequence; each one ends with a reviewed deliverable before the next begins.
- Federated training is simulated on one machine: sensors are split into 3 to 5 clients, and only model weights are shared.
- Every experiment is repeated in IID and non IID settings and compared with centralized and local-only baselines.
- Results are judged on MAE, RMSE, and communication bytes per round and convergence.

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[2] S. Kaur, S. S. Sehra and D. Ebrahimi, FedLLM: A Privacy-Preserving Federated Large Language Model for Explainable Traffic Flow Prediction, arXiv:2604.16612, 2026.

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[4] C. Abboud and S. Khalil, FedTraffic: A Hierarchical Federated Learning Framework for Traffic Flow Prediction in Intelligent Transportation Systems, Eng, vol. 7, no. 8, art. 362, 2026.

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[6] J. Feng, C. Du and Q. Mu, Traffic Flow Prediction Based on Federated Learning and Spatio-Temporal Graph Neural Networks, ISPRS International Journal of Geo-Information, vol. 13, art. 210, 2024.

[7] S. Kaur, S. S. Sehra, D. Ebrahimi and E. A. Mohammed, A Traffic-Aware Federated Learning Prediction Framework with Custom Aggregation, Machine Learning with Applications, vol. 23, art. 100829, 2026.

[8] X. Gou et al., XTraffic: A Dataset Where Traffic Meets Incidents with Explainability and More, arXiv:2407.11477, 2024. Data at https://xaitraffic.github.io.

Thank You for Your Attention!